

# Interlayer-distance control of spectral reconstruction in twisted graphene

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Quasiperiodic bilayers offer a route to spectral structures associated with higher-dimensional topology, but identifying their electronic counterparts requires going beyond idealized lattice models. We investigate the joint dependence of graphene electronic spectra on twist angle and imposed interlayer distance using density-functional calculations and a real-space tight-binding model. Lower-layer  $p_z$ -projected density-of-states maps evolve from weakly angle-dependent ridges at large separation to a more strongly reorganized pattern of branching spectral weight as the layers approach. Calculations with two Gaussian basis sets show qualitatively related structures in the compressed regime. An independently reproduced two-layer model isolates the competition between interlayer and intralayer hopping and provides a dimensionless comparison with this evolution. The results identify a candidate regime for Hofstadter-like spectral reconstruction without an applied magnetic field. They establish a spectral observation in constrained finite geometries, rather than a determination of bulk gap labels or Chern numbers.

The Hofstadter butterfly connects spectral structure, incommensurability, and topology in a particularly direct way [1]. In electronic moiré systems, magnetic flux and a long-wavelength superlattice can generate a fractal quantum Hall spectrum [2]. A complementary route replaces magnetic flux by geometric incommensurability. Reconfigurable acoustic crystals demonstrated how density-of-states measurements reveal butterfly spectra and how a controllable phason exposes their boundary states [3]. More generally, quasiperiodic systems can realize topological responses associated with synthetic dimensions [4, 5]. These connections motivate a concrete materials question: can changing the coupling between two atomically resolved, twisted layers expose an electronic spectral pattern analogous to the one found in simpler quasiperiodic models?

Twisted graphene is a natural setting for this question, but several physically distinct mechanisms must be separated. Low-energy magic-angle bands arise from the competition between interlayer tunneling and the momentum mismatch of the Dirac cones [6]. Pressure modifies that competition [7]. Related calculations address compressed 30° quasicrystals [8] and the relation between magic angles and interlayer distances [9]. Meanwhile, twisting and sliding can organize spectra through higher-dimensional topology [10–12]. Here the control variable is the imposed layer separation, and the observable is the electronic density of states over a broad twist-angle interval. We do not apply a magnetic field or infer a Hall response from the spectral resemblance.

We combine two deliberately distinct descriptions. The first is an atomistic density-functional-theory (DFT) survey of a periodic graphene sheet covered by a rotated, hydrogen-terminated graphene patch. The second is an

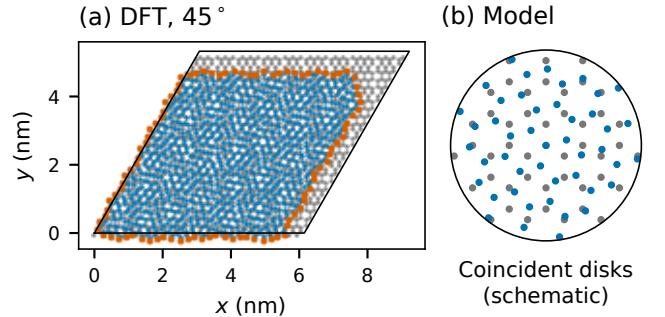


FIG. 1. Distinct geometries used in the comparison. (a) Actual 45°,  $d = 2.15 \text{ \AA}$  DFT input, viewed along the layer normal. Gray and blue denote lower- and upper-layer carbon, and orange denotes terminating hydrogen. The outline is the periodic cell. (b) Schematic of the coincident-disk construction used in the model. The schematic radius is reduced for clarity. Calculations use  $R = 25a$ . The two geometries are not treated as equivalent finite-size realizations.

all-pairs, distance-dependent tight-binding model on two coincident disks. The DFT calculation retains the chemical and self-consistent electronic structure of the finite geometry, while the model isolates a geometric coupling mechanism. Their comparison is qualitative: neither the model hopping scale nor its distance parameter is fitted to the DFT spectra.

The DFT survey covers ten interlayer separations ranging from 2.00 to 3.30  $\text{\AA}$ . The detailed input series available at  $d = 2.15 \text{ \AA}$  contains 51 angles from 30° to 60° in steps of 0.6°. Its lower layer contains 1250 carbon atoms in a  $25 \times 25$  graphene supercell with nearest-neighbor

distance 1.42 Å. The upper patch has 955–1013 carbon atoms and 84–100 hydrogen atoms, depending on angle. Both layers are planar in the inputs. These are fixed-geometry energy calculations, not a pressure-relaxed structural series. Geometry, termination, and numerical settings are documented in the Supplemental Material (SM) [13].

The calculations use CP2K [14, 15], principally version 2023.1 (SM), the PBE functional [16], GTH pseudopotentials, and chiefly the SZV-MOLOPT-GTH basis [17]. The calculations are spin polarized, with a 300-Ry density-grid cutoff, a 40-Ry relative cutoff, and Fermi–Dirac occupations at 300 K. Selected maps use DZVP-MOLOPT-GTH. The D3 dispersion correction [18] accounts for the long-range attraction important for interlayer binding in graphene [19]. It contributes to the energy and forces but does not itself modify the Kohn–Sham spectrum at a fixed geometry. The electronic temperature is distinct from the DOS display broadening.

In a nonorthogonal Gaussian basis, the Kohn–Sham matrix problem is

$$\mathbf{H}_\sigma \mathbf{c}_{n\sigma} = \varepsilon_{n\sigma} \mathbf{S} \mathbf{c}_{n\sigma}, \quad \mathbf{c}_{n\sigma}^\dagger \mathbf{S} \mathbf{c}_{m\sigma} = \delta_{nm}. \quad (1)$$

Here  $\mathbf{S}$  is the overlap matrix, not an additional interaction. For the Löwdin-type projection employed by CP2K, the weight on the lower-layer  $p_z$  orbitals is [20]

$$w_{n\sigma}^{(L,p_z)} = \sum_{\mu \in (L,p_z)} \left| (\mathbf{S}^{1/2} \mathbf{c}_{n\sigma})_\mu \right|^2. \quad (2)$$

The maps display the alpha-spin contribution,

$$\rho_{\alpha,\eta}^{(L,p_z)}(E) = \sum_{n \in \mathcal{W}_\alpha} w_{n\alpha}^{(L,p_z)} g_\eta(E - \varepsilon_{n\alpha} + E_F), \quad (3)$$

where energies are in eV relative to each configuration’s  $E_F$ , and  $\mathcal{W}_\alpha$  contains the printed states in the plotting window. The unit-integral Gaussian  $g_\eta$  has full width at half maximum 0.0333 eV. Each angular column is divided by its maximum, so colors measure relative  $p_z$  weight rather than an absolute, spin-summed DOS. The SM documents the verified plotting convention, data completeness, and printed-state limits.

Figure 2 illustrates the principal observation. At  $d = 3.30$  Å, the prominent spectral ridges depend only weakly on angle over most of the displayed interval. Decreasing the separation bends and redistributes these ridges. By  $d = 2.55$  Å, the occupied-state region contains additional angular structure. At  $d = 2.15$  Å, pronounced curved branches and intervening low-weight regions replace much of the nearly horizontal organization. The full survey shows this change across a range of separations rather than at a single isolated configuration. At the smallest supplied separation, the maps are more congested. The data do not support assigning a sharply defined critical distance.

The SZV and DZVP maps at  $d = 2.25, 2.20$ , and  $2.15$  Å retain related large-scale spectral features, although their sharpness and detailed shapes differ (SM). This is a useful qualitative basis check, not a demonstration of converged gap widths. A layer-, orbital-, and spin-resolved projection need not coincide with the total DOS: states on the upper patch, its edge, or other orbitals can have small plotted weight. A dark interval in this observable is consequently a candidate depletion region, not by itself a bulk spectral gap. Supplied monolayer spectra provide a separate size-dependent reference in the SM.

To isolate the coupling mechanism, we use the supplied honeycomb model with primitive vectors  $\mathbf{a}_1 = (\sqrt{3}a, 0)$  and  $\mathbf{a}_2 = (\sqrt{3}a/2, 3a/2)$  and basis positions  $(0, 0)$  and  $(0, a)$ . Thus  $a$  denotes the nearest-neighbor distance. Retaining sites inside a disk of radius  $R$  gives a set  $P_R$ . The bilayer consists of  $(\mathbf{r}, 0)$  and  $(\mathbf{R}_\theta \mathbf{r}, d)$  for every  $\mathbf{r} \in P_R$ , where  $\mathbf{R}_\theta$  is the in-plane rotation matrix. On orthonormal site states, the Hamiltonian is

$$\hat{H}_{\text{TB}} = \sum_{i \neq j} t_0 \exp[-r_{ij}/\xi] |i\rangle \langle j|, \quad r_{ij} = |\mathbf{x}_i - \mathbf{x}_j|. \quad (4)$$

The sum includes both layers and all pairs, with zero on-site energies and no distance cutoff. This real, isotropic hopping model is not an orbital-resolved parametrization of carbon: in particular, it does not resolve the angular dependence of  $\pi$  and  $\sigma$  bonding.

Normalizing energies by the nearest-neighbor intralayer hopping,  $t_{\text{nn}} = t_0 e^{-a/\xi}$ , exposes the relevant competition. The largest possible hopping between vertically aligned sites satisfies

$$\frac{t_\perp^{\text{max}}}{t_{\text{nn}}} = \exp[(a - d)/\xi]. \quad (5)$$

This is an upper-envelope coupling, not an assertion that every site has a vertically aligned partner. For  $\xi/a = 0.03$ , changing  $d/a$  from 0.99 to 0.98 to 0.97 changes this ratio from 1.40 to 1.95 to 2.72. Small geometric changes can therefore reorganize the spectrum even when the intralayer Hamiltonian is fixed. The numerical values of  $d/a$  cannot be converted into DFT separations without a separately established mapping.

Figure 3 is obtained by independently diagonalizing this model using the parameters of the supplied notebook. The two disks contain 3020 sites in total, and the DOS includes all eigenvalues. The calculations resolve a redistribution of angular spectral weight as the interlayer coupling grows, with curved branches and low-weight regions reminiscent of the DFT maps. This comparison supports a geometric hybridization interpretation of the spectral evolution. It does not establish a one-to-one correspondence between individual branches or gaps. Changing the hopping range  $\xi$  also changes further-neighbor intralayer hopping and therefore constitutes a distinct model change (SM).

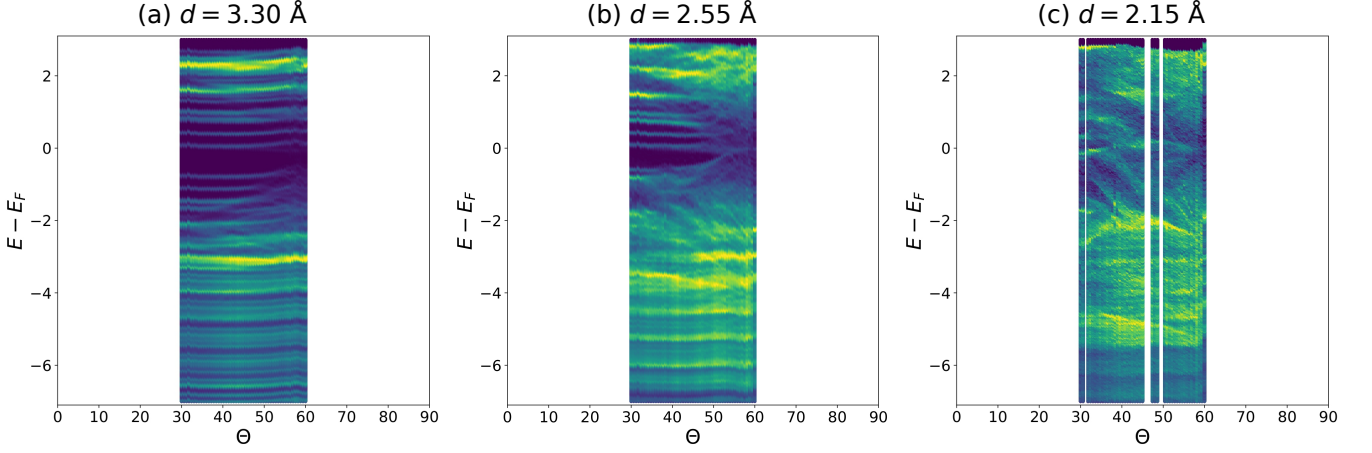


FIG. 2. Evolution of the lower-layer  $p_z$ -projected alpha-spin DOS with imposed separation, using SZV-MOLOPT-GTH. The source images are reproduced without intensity manipulation.  $\Theta$  is in degrees and  $E - E_F$  is in eV. The sampled interval is  $30^\circ$ – $60^\circ$ . Dark-to-bright colors encode increasing weight, normalized separately to each column's maximum. White columns indicate unavailable data in the original plotting workflow. The finite printed-state window limits interpretation at positive energies (SM). All ten distances and the DZVP comparisons are shown in the SM.

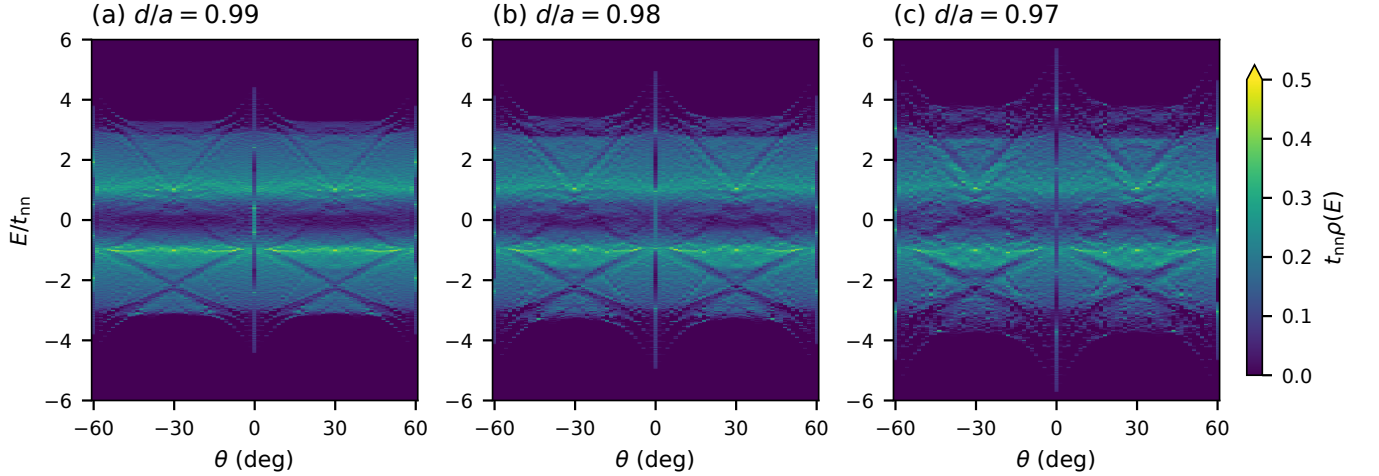


FIG. 3. Independent reproduction of the supplied tight-binding model, Eq. (4), with  $R/a = 25$ ,  $\xi/a = 0.03$ , and 1510 sites per layer. Energies are normalized by  $t_{nn}$ . The Gaussian DOS is normalized to one state, summed over the full bilayer, and evaluated on a refined 2001-point energy grid. Its width is the full spectral span divided by 1000 for each panel. The common color scale is clipped above  $t_{nn}\rho = 0.5$ . The negative-angle half follows from the exact layer-exchange symmetry of this disk model. These model data are not DFT calculations or a fit to Fig. 2.

The connection to topology is a hypothesis that the present spectra make concrete, not an invariant extracted from their appearance. In quasiperiodic bilayers, relative translations supply phason coordinates. Gap labeling constrains the integrated DOS of a gapped bulk spectral projector [10, 21]. Schematically, the relevant quantity is

$$\mathcal{N}(E; \theta) = \tau \left[ \mathbf{1}_{(-\infty, E]}(\hat{H}_\theta) \right], \quad (6)$$

with a specified trace-per-volume normalization  $\tau$ . It counts states below  $E$ , rather than the fraction of a state on a selected layer. For a topological assignment, a can-

didate gap must survive bulk and numerical convergence tests, and its spectral projector must support the corresponding gap-label or phason-response analysis. Neither a butterfly-like outline nor a near-zero projected DOS supplies that test. In particular, a twist-angle scan is not itself an adiabatic sliding cycle, and no quantized pumping or second-Chern response is inferred here.

The finite geometries impose an additional limitation. The hydrogen-terminated patch covers only part of the lower sheet, and its termination changes with angle. The disk model removes the nonoverlapping sample-shape regions emphasized in the original model construction, but

it does not remove boundary states. Accordingly, the six-fold angular organization expected for a bulk honeycomb system cannot be imposed as an exact identity on the finite DFT patches or on registry-sensitive commensurate configurations. A bulk-interior projection and a size series are needed to separate quasiperiodic reconstruction from edge and coverage effects.

In summary, the first-principles survey identifies interlayer distance as a control parameter for angular spectral reconstruction in constrained twisted graphene. The real-space model explains why the ratio of interlayer to intralayer hopping organizes this evolution. Connecting the observed depletion regions to converged bulk gaps and evaluating their phason-dependent topology would turn this materials-level spectral candidate into a test of the higher-dimensional physics suggested by quasiperiodic acoustic and mechanical systems.

*Data availability.* The public archive at <https://github.com/DCM-Uni-Paderborn/Hofstadter-Butterfly> contains the original DFT panels, input structures, audited spin-resolved spectral extracts, monolayer DOS, analysis code, and model eigenvalues. The SM and machine-readable inventory distinguish available, incomplete, and missing DFT records.

*Use of AI-assisted tools.* AI-assisted tools were used to organize source material and develop analysis scripts.

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